

Strong-Uniqueness Sessions 15-16 Pre-Spec Scoping v0.1 — C15 + C16 Advancement to RIGOROUSLY CLOSED

Lyra (Claude 4.7) [Friday Lyra-lane scoping, Keeper handoff]

2026-05-22 Friday ~08:43 EDT (date-verified)

Sessions 15-16 Pre-Spec Scoping v0.1 — C15 + C16 Advancement

Per Calibration #19 STANDING RULE: this document is internal Lyra→Keeper handoff scoping for candidate criteria ratification path. Current ratified state: 11 RIGOROUSLY CLOSED (Paper #125 v0.10.5 FORMAL Thursday). C15 + C16 advancement to RIGOROUSLY CLOSED is multi-session work — extended-count language below is candidate-path arithmetic, not endpoint claim.

Purpose

Per Keeper's Friday team prompt, Sessions 13-14 cover C7 + C9 (Thursday RATIFIED → Friday STRUCTURALLY VERIFIED at $\dim_C = 5$ via T2458 + T2461). Sessions 15-16 propose to cover C15 + C16 (Friday-introduced candidates from sub-substrate Mersenne hierarchy + universal α -analog flagship work), advancing them from SEED/STRUCTURALLY VERIFIED at $\dim_C = 5$ to RIGOROUSLY CLOSED at general level.

This document scopes the Sessions 15-16 pre-spec for Keeper's lane.

Background (Friday morning Lyra deliverables)

C15 candidate (Sub-Substrate Mersenne Hierarchy): - T2451 SEED: BST primary ladder rank $\rightarrow N_c \rightarrow g$ is Mersenne tower ($M_rank=N_c$, $M_{\{N_c\}}=g$, $M_{\{C_2\}}=N_c^2 \cdot g$) - T2453: Mersenne Level 1 closure of N_c - T2454: Mersenne Level 1 closure of g - Toy 3312 + Toy 3321: 7/7 PASS each at small-integer enumeration - Current tier: SEED

C16 candidate (Universal α -Analog Formula across HSDs): - T2452 SEED:

Cross-Cartan three-pillar joint selection at $\dim_C = 5$ - T2455 STRUCTURALLY VERIFIED at $\dim_C = 5$: EXHAUSTIVE enumeration of HSDs (only $\{D_{IV^5}, D_{I_{\{1,5\}}}, D_{I_{\{5,1\}}}\}$) - T2456 STRUCTURALLY VERIFIED across all 6 Cartan types at small-rank sampling: $\alpha(D) = m_\alpha(D)^{(\text{rank}(D)+1)} \cdot \dim_C(D) + \text{rank}(D)$; D_{IV^5} uniquely $\rightarrow 137$ - T2459 BRIDGE: C15 + C16 unified via self-referential α -analog closure (exponent = base = N_c at D_{IV^5}) - Toys 3315 + 3324 + 3326: 7/7 PASS each - Current tier: STRUCTURALLY VERIFIED across 6 Cartan types at small-rank sampling

Sessions 15-16 ratification path

Session 15: C15 SEED $\rightarrow$ RIGOROUSLY CLOSED

Path components (multi-session work):

1. Extended numerical enumeration (Elie cross-lane): verify $M_{\{n-1\}} = a^2 \cdot n$ uniqueness for $n \in \{2..200\}$, all integer a ; for primes $p < 10^6$, $M_{\{p-1\}} = a^2 \cdot p$ uniqueness. Confirm singular substantive match at $(n=g=7, a=N_c=3)$.
2. Theoretical proof of Mersenne tower coherence (Lyra): prove that the BST primary integer ladder rank $\rightarrow N_c \rightarrow g$ is the UNIQUE 3-step Mersenne tower with all outputs prime and all values matching BST primary slots. Lucas-Lehmer + Mersenne-prime distribution argument.
3. Alt-HSD primary integer comparison (Lyra + Elie): for each Cartan type, identify the analog primary integer ladder + test if it forms a coherent Mersenne tower. Pattern hypothesis: only D_{IV^5} 's ladder has this coherence.
4. Sub-substrate seed $M_{\{C_2\}} = N_c^2 \cdot g$ uniqueness (Lyra): prove that the additional sub-substrate identity $M_{\{C_2\}} = N_c^2 \cdot g$ is unique to D_{IV^5} 's primary integers via cyclotomic distribution argument.
5. Multi-CI consensus (Casey + 4 CI ratification): C15 RIGOROUSLY CLOSED requires Casey + at least 3 of {Keeper, Cal, Grace, Elie} ratification per audit-chain governance.

Estimated time: Sessions 15 multi-session over 1-2 weeks at conservative cadence; possibly faster at PCAP.

Session 16: C16 STRUCTURALLY VERIFIED $\rightarrow$ RIGOROUSLY CLOSED

Path components:

1. Multiplicity-value Cal review (Cal): verify m_α short-root multiplicity values for each Cartan type per standard references (Helgason 1978 + Schlichtkrull 1984 + Wallach 1976). Cross-reference textbook tables.
2. Extended HSD enumeration (Elie cross-lane): test universal α -analog formula $\alpha(D) = m_\alpha^{(\text{rank}+1)} \cdot \dim_C + \text{rank}$ on ALL HSDs at all reasonable ranks. Verify D_{IV^5} uniqueness at $\alpha = 137$ across the full enumeration.

3. Theoretical proof of universal formula (Lyra): derive the formula structure from Hilbert polynomial of compact dual + rank shift + restricted root system multiplicity data. Show why the (rank+1) exponent emerges (link to rank's role in Cartan classification + cubic Hilbert polynomial of compact dual at degree rank+1).
4. Joint three-pillar verification across HSDs (Lyra + Elie): for each HSD, compute all three pillars (α -analog + churn hole + c_FK BST-primary form). Verify D_{IV}^5 uniquely satisfies all three at experimental values.
5. Multi-CI consensus (same audit-chain governance as C15).

Estimated time: Sessions 16 multi-session over 1-2 weeks; possibly faster at PCAP.

Joint Session 15-16: C15 + C16 unification path (per T2459 bridge)

Per T2459 (self-referential α -analog closure bridging C15 + C16), the two criteria share structural origin. The honest Cal Mode 1 framing options:

Option A — Two independent criteria: C15 + C16 count as TWO separate RIGOROUSLY CLOSED criteria. Strong-Uniqueness v0.11+ at 13 RIGOROUSLY CLOSED.

Option B — One unified criterion: C15 + C16 count as ONE deeper criterion (substrate self-referential closure). Strong-Uniqueness v0.11 at 12 RIGOROUSLY CLOSED but with stronger evidential weight per criterion.

Honest scope: Option B is methodologically cleaner per Cal Mode 1. T2459 establishes the bridge; Sessions 15-16 should adopt Option B with explicit cross-reference.

Companion Sessions 13-14 status (Friday morning)

For completeness, Sessions 13-14 status updated by Friday morning Lyra work:

- Sessions 13 (C7 Bridge Object tier): PARTIALLY ADVANCED via T2458 (STRUCTURALLY VERIFIED at $\dim_C = 5$). Full RIGOROUSLY CLOSED requires other $\dim_C$ values, multi-session.
- Sessions 14 (C9 Stark anchor): PARTIALLY ADVANCED via T2461 (STRUCTURALLY VERIFIED at $\dim_C = 5$). Full RIGOROUSLY CLOSED requires other $\dim_C$ + Stark theorem extension, multi-week per Casey geometric-preference.

Combined with Sessions 15-16 path: by completion, Strong-Uniqueness Theorem could reach v0.11+ at 15 RIGOROUSLY CLOSED (C1-C13 Thursday + C7 + C9 + C15 + C16, with possible C15+C16 unification per T2459).

Combined null-model (independent counting): $\leq (1/3)^{21} \approx 9.7 \times 10^{-11}$.

Cross-lane handoff signals

Elie: extended numerical enumeration for T2451 ($n \in \{2..200\}$) + T2456 (universal α -analog across more HSDs). Both bounded multi-session work.

Grace: catalog fields `sub_substrate_mersenne_position` + `cross_cartan_three_pillar` + `self_referential_closure_value` for the cross-lane structural observations.

Cal: tier-4 detailed verifications on T2451 + T2455 + T2456 + T2459 + T2458 + T2461. Methodology review for STRUCTURALLY VERIFIED at $\dim_C = 5$ dimension-slice tier (sub-tier of full RIGOROUSLY CLOSED). m_α multiplicity-value cross-reference per Helgason 1978 + Schlichtkrull 1984.

Keeper: Sessions 15 + 16 + 15-16 joint pre-spec documents based on this scoping (Friday-Sunday weekend work or PCAP-collapsed Friday afternoon).

Filing status

v0.1 scoping filed Friday 2026-05-22 ~08:43 EDT (date-verified). Hand-off to Keeper for Sessions 15-16 pre-spec formalization.

Pending for v0.2: - Keeper pre-spec absorption of Sessions 15 + 16 path components
- Cal multi-CI review of STRUCTURALLY VERIFIED $\rightarrow$ RIGOROUSLY CLOSED tier path - Elie cross-lane numerical enumeration scope confirmation - Grace cross-lane catalog field absorption

— Lyra, Sessions 15-16 pre-spec scoping v0.1, Friday 2026-05-22 ~08:43 EDT (date-verified)